PROJECT REPORT

Intern Support AI Chatbot

An NLP-Based FAQ Assistant for Internship Support

Submitted by

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1. Executive Summary

The Intern Support AI Chatbot is a Python-based web application designed to help interns find answers to common internship-related questions. It provides information about task submissions, deadlines, supervisors, certificates, GitHub, Python, project documentation, and technical issues.

The chatbot uses Natural Language Processing (NLP) techniques to compare users' questions with questions stored in a Frequently Asked Questions (FAQ) dataset. It uses Term Frequency-Inverse Document Frequency (TF-IDF) to convert text into numerical representations and cosine similarity to identify the most relevant question.

The application is developed using Streamlit, which provides an interactive chat interface. When a suitable FAQ match is found, the chatbot displays the associated answer. If a question is unrelated or no sufficiently similar question is found, the chatbot provides a fallback response directing the user to their internship supervisor.

Initial testing indicates that the chatbot can answer many common internship questions and appropriately handle unrelated queries. Some question-matching cases may require further refinement to improve accuracy.

2. Introduction

Interns frequently require guidance regarding internship policies, task submissions, deadlines, certificates, supervisors, programming tools, and project requirements. Finding answers to these questions can take time, particularly when supervisors receive repeated queries about common issues.

The Intern Support AI Chatbot addresses this problem by providing a simple, interactive interface through which interns can ask questions and receive answers from a predefined FAQ dataset.

The system uses text similarity rather than generating new answers independently. This approach keeps responses consistent with the information stored in the dataset and makes the application suitable for providing basic internship guidance.

3. Problem Statement

Interns may experience difficulties obtaining quick answers to routine questions about their internship activities. Depending entirely on supervisors for every small query can delay progress and increase the number of repetitive support requests.

A lightweight, automated support system can help interns access commonly required information quickly. The proposed chatbot uses NLP-based question matching to identify relevant FAQ entries and display their corresponding answers.

4. Project Objectives

The main objectives of the project are:

1. To develop an interactive chatbot for answering common internship-related questions.

2. To use NLP techniques to match users' questions with relevant FAQ entries.
3. To provide information about submissions, deadlines, certificates, GitHub, Python, and technical support.
4. To support alternative phrasings of common questions.
5. To provide fallback responses when a suitable answer cannot be identified.
6. To develop a simple, accessible interface using Streamlit.
7. To demonstrate the practical application of Python and NLP techniques in an internship support system.

5. Project Scope

The chatbot focuses on answering general questions related to internship activities and technical tasks. Its scope includes:

- Internship timings and meetings.
- Task submission procedures.
- Deadlines and extension requests.
- Supervisor communication.
- Internship certificates and completion requirements.
- GitHub repositories and code uploads.
- Python installation and troubleshooting.
- Project README documentation.
- Technical issues and submission problems.
- Portfolio use and project confidentiality.

Note: The chatbot provides general guidance based on its FAQ dataset. Official decisions regarding internship policies, deadlines, certificates, and eligibility must still be confirmed with the relevant coordinator or supervisor.

6. Technologies Used

6.1 Python

Python is the main programming language used to develop the chatbot. It handles data processing, text preparation, question matching, and application logic.

6.2 Streamlit

Streamlit is used to build the chatbot's web interface. It allows users to enter questions, view responses, maintain chat history, and clear the conversation.

6.3 Pandas

Pandas is used to load and process the FAQ dataset stored in CSV format.

6.4 Scikit-learn

Scikit-learn provides the text vectorization and similarity-computation tools used by the chatbot.

6.5 TF-IDF Vectorization

TF-IDF converts text questions into numerical vectors based on the importance of words and word combinations within the available question collection. This representation allows the system to compare a user's question with the stored FAQ questions.

6.6 Cosine Similarity

Cosine similarity measures the similarity between numerical text vectors. The chatbot uses it to identify the stored question that most closely matches the user's input.

6.7 CSV Dataset

The `intern_faq.csv` file stores the predefined questions and answers used by the chatbot.

7. Dataset Description

The project uses a structured FAQ dataset named `intern_faq.csv`. The dataset contains 30 question-and-answer pairs and has two columns:

Column	Description
`question`	Contains a frequently asked internship question.
`answer`	Contains the corresponding response.

The dataset covers different areas of internship support, including technical troubleshooting, project submissions, supervisor communication, certificate requirements, and GitHub usage.

The chatbot uses these stored answers as its knowledge source. Consequently, the quality of its responses depends on the accuracy, completeness, and relevance of the dataset.

8. System Architecture and Working

The chatbot follows an NLP-based information-retrieval approach. The workflow consists of eight steps:

Step	Stage	Description
1	User Input	The user enters an internship-related question through the Streamlit chat interface. Example: "How can I upload my project to GitHub?"
2	Text Preprocessing	The question is converted to lowercase, punctuation is removed, and whitespace is standardized. This reduces differences caused by text formatting.

Step	Stage	Description
3	FAQ Preparation	Questions and answers are loaded from the CSV dataset. Alternative phrasings for supported questions are also used to improve matching when users express the same intent differently.
4	TF-IDF Vectorization	The questions are converted into numerical vectors using TF-IDF, considering individual words and two-word combinations.
5	Similarity Calculation	Cosine similarity is calculated between the user's question and the stored questions, including their alternative phrasings.
6	Best-Match Selection	The candidate question with the highest similarity score is selected and its FAQ answer is identified.
7	Threshold Check	A minimum similarity threshold of 0.30 is applied. If the best score is below it, a fallback response is returned instead of an insufficiently similar FAQ.
8	Response Display	The selected answer or fallback message is displayed in the Streamlit chat interface.

9. Main Features

9.1 Interactive Chat Interface

Users can enter questions and receive responses through a browser-based chat interface.

9.2 FAQ-Based Question Answering

The chatbot retrieves answers from the predefined FAQ dataset.

9.3 NLP-Based Question Matching

TF-IDF and cosine similarity help the system match questions based on textual similarity.

9.4 Alternative Question Phrasings

The application includes alternative phrasings for supported questions to improve its ability to handle variations in user input.

9.5 Fallback Response

When the system cannot identify a sufficiently similar FAQ, it displays a fallback message asking the user to contact their internship supervisor.

9.6 Chat History

The interface displays the conversation so users can review the questions and responses exchanged during the session.

9.7 Clear Chat Option

Users can clear the current conversation through the interface.

10. Project Structure

The project directory is organized as follows:

```
Intern-Support-AI-Chatbot/
|
|-- app.py
|-- intern_faq.csv
|-- requirements.txt
|-- .gitignore
|-- README.md
`-- .venv/
```

File	Description
`app.py`	Streamlit application, text preprocessing, question-matching logic, similarity threshold, and response handling.
`intern_faq.csv`	Stores the FAQ questions and answers.
`requirements.txt`	Lists the Python packages required to run the application.
`.gitignore`	Prevents local files and folders, such as the virtual environment and Python cache, from being tracked by Git.
`README.md`	Project overview, installation instructions, features, and usage information.
`.venv/`	Local Python virtual environment. It should not be uploaded to GitHub.

11. Application Setup and Execution

The application can be run locally on a computer with Python installed.

Step 1: Open the Project Directory

Open the project folder in Visual Studio Code.

Step 2: Create a Virtual Environment

If a virtual environment does not already exist, run:

```
python -m venv .venv
```

Step 3: Activate the Virtual Environment

On Windows PowerShell, run:

```
.\.venv\Scripts\Activate.ps1
```

Step 4: Install Dependencies

Run:

```
python -m pip install -r requirements.txt
```

Step 5: Launch the Application

Run:

```
python -m streamlit run app.py
```

Streamlit will start the application and provide a local address that can be opened in a web browser.

12. Testing and Results

Manual testing was conducted using different types of questions to evaluate the chatbot's responses. The tests included common FAQ questions, alternative question phrasings, technical-support questions, and unrelated general-knowledge questions.

12.1 Test Cases

Test Question	Expected Behaviour	Observed Result
How can I obtain my certificate?	Return certificate-related information.	**Passed**
Who should I contact about certificate eligibility?	Direct the user to the internship coordinator.	**Passed**
I am having trouble with Python. What should I do?	Return Python troubleshooting guidance.	**Passed**
How do I upload my code to GitHub?	Return GitHub upload instructions.	**Passed**
How can I turn in my assignment?	Return task submission instructions.	**Passed**
Are internship meetings compulsory?	Return meeting policy guidance.	**Passed**
What should I include in my README file?	Explain the expected README contents.	**Passed**
What is the capital of France?	Return the fallback response.	**Passed**
What is artificial intelligence?	Return the fallback response.	**Passed**
I need extra time to finish my assignment. What should I do?	Return the extension-request answer.	**Needs improvement**

12.2 Testing Observations

The chatbot successfully answered several common internship questions, including certificate requests, GitHub usage, task submissions, technical troubleshooting, and supervisor communication. It also returned the fallback response for unrelated questions such as asking about the capital of France or the definition of artificial intelligence.

One test involving an assignment extension returned a response about an incomplete task rather than the more relevant extension-request answer. This indicates that similar questions can sometimes be confused by text-similarity matching.

In the shared test batch, 16 out of 17 questions were assessed as suitable, giving an observed result of approximately 94.1% for that small sample.

Note: This figure represents only the manually tested examples. It should not be considered a formal or general accuracy measurement of the complete chatbot.

13. Limitations

Although the chatbot performs well on many tested examples, it has several limitations:

1. **Limited Knowledge:** The chatbot can answer only questions covered by its FAQ dataset and supported alternative phrasings.
2. **Similarity-Based Matching:** Text similarity may confuse questions that use similar words but express different intentions.
3. **Predefined Answers:** The chatbot retrieves existing answers rather than generating new, context-aware responses.
4. **Threshold Dependence:** The selected similarity threshold may not perform equally well for every type of question.
5. **Dataset Dependence:** Missing, outdated, or unclear FAQ entries can reduce answer quality.
6. **Policy Limitations:** The chatbot cannot independently verify official internship policies or make decisions about certificate eligibility.

14. Future Improvements

Several improvements could be introduced in future versions:

- Expand the FAQ dataset with additional internship-related questions.
- Improve matching for similar requests, such as deadline extensions and incomplete tasks.
- Create a larger test set with predefined expected answers.
- Measure performance using systematic evaluation metrics.
- Add multilingual support for users who prefer Urdu or other languages.
- Introduce a feedback feature so users can report incorrect or unhelpful answers.
- Add more internship support categories.
- Deploy the application online after reviewing security and privacy requirements.
- Consider more advanced NLP methods if they are needed for better semantic understanding.

15. Conclusion

The Intern Support AI Chatbot demonstrates the practical use of Natural Language Processing techniques to provide automated answers to common internship-related questions.

The application combines Python, Streamlit, Pandas, and Scikit-learn to create an interactive FAQ assistant. TF-IDF vectorization and cosine similarity allow the chatbot to compare user questions with stored FAQ entries, while alternative phrasings improve matching for common variations. A fallback mechanism helps prevent the system from returning answers when no sufficiently similar FAQ is found.

Initial manual testing demonstrated useful performance on several common internship questions and appropriate fallback behaviour for unrelated queries. The testing also identified a question-matching case that could be improved.

Overall, the project provides practical experience in Python development, NLP-based text processing, dataset management, web application development, and software testing. It can serve as a foundation for a more advanced internship support system.

Appendix: Example User Questions

The following questions can be used to demonstrate the chatbot:

1. How can I obtain my certificate?
2. How do I submit my task?
3. Who should I contact about certificate eligibility?
4. How do I upload my code to GitHub?
5. What should I do if my Python code is not working?
6. How can I request an extension?
7. What should I include in my README file?
8. Can I use Visual Studio Code for my tasks?
9. What if my supervisor does not respond?
10. What is the capital of France?

These questions demonstrate the chatbot's FAQ retrieval, alternative question matching, and fallback behaviour.